Design of brain-simulating liquid and numeric phantoms for evaluating hemodynamic changes in intraoperative functional brain mapping studies

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ABSTRACT

The optical imaging described here is a marker-free, contactless, and non-invasive technique that is able to monitor hemodynamic brain response following neuronal activation during neurosurgery. However, a robust quantification is complicated to perform during neurosurgery due the critical context of the operating room, which makes the calibration and adjustment of optical devices more complex. To overcome this issue, tissue-simulating objects that mimic the properties of biological tissues are required for the development of detection or diagnostic imaging systems. In this study, we evaluated the performance of quantification of chromophore concentration changes measured by experimental setups using two phantoms: a liquid and a numeric brain-simulating phantom. These phantoms mimicked an exposed cerebral cortex as well as the slow concentration changes that occur after neuronal stimulation and the periodic changes due to heartbeat.

Keywords: liquid phantom, numeric phantom, brain-simulating phantoms, neuroimaging, optical imaging

1. INTRODUCTION

Non-invasive functional brain mapping is an imaging technique that allows the location of functional areas of the patient's brain. This technique is used during brain tumour resection surgery to indicate to the neurosurgeon the cortical tissues which should not be removed without cognitive impairment. Optical imaging is a marker-free, contact-less, and non-invasive technique especially adapted for intraoperative functional brain mapping applications. Optical devices such as RGB cameras combined with a white light illumination allow the analysis of the light absorption to monitor the brain activity with quantification of the concentration changes in oxy- (ΔC_{HbO_2}) and deoxygenated hemoglobin (ΔC_{Hb}) in brain cortex.[1–3]

A robust quantification of these biomarkers is complicated to perform during neurosurgery due the critical context of the operating room, which makes the calibration and adjustment of optical devices more complex. To overcome this issue, tissue-simulating objects that mimic the properties of human or animal tissues are required for the development of detection or diagnostic imaging systems. These so-called "phantoms" may be used to evaluate, optimise, compare or control the non-regression of imaging systems.[4–6] To model biological tissues, scattering and absorbing materials can be used. An exhaustive review of the different types of phantoms can be found here.[4]

In this study, hemodynamic events that occurs during cerebral activities were modelled with a numeric brain phantom based on Monte Carlo simulations[7,8] and a liquid phantom with adjustable optical properties. For both phantoms, we modelled the slow concentration changes occurring after neuronal stimulation and the periodic changes due to heartbeat. The modified Beer-Lambert law was then used to evaluate the quantification performance of chromophore concentration changes in both phantoms.

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2. MATERIAL AND METHODS

2.1 Modified Beer Lambert law

When using a spectral acquisition device having K spectral channels, the modified Beer-Lambert law can be used to quantify N chromophores' concentration changes over time:

$$\begin{bmatrix} \Delta A_1(t) \\ \vdots \\ \Delta A_K(t) \end{bmatrix} = \begin{bmatrix} E_{1,1} & \dots & E_{1,N} \\ \vdots & \ddots & \vdots \\ E_{K,1} & \dots & E_{K,N} \end{bmatrix} \times \begin{bmatrix} \Delta C_1(t) \\ \vdots \\ \Delta C_N(t) \end{bmatrix} \tag{1}$$

with

$$E_{i,n} = \int \epsilon_n(\lambda).D_i(\lambda).S(\lambda).L(\lambda).d\lambda. \tag{2}$$

$\Delta A_i(t)$ is the absorbance changes measured at time t:

$$\Delta A_i(t) = \log_{10}\left(\frac{R_i^0}{R_i(t)}\right), \tag{3}$$

where $R_i(t)$ is the reflection intensity measured at time t by the spectral channel i ($i \in [1;K]$), R_i^0 is the reference reflection intensity measured by the spectral channel i. Reference intensities were collected at the beginning of the acquisition, during a "rest" period (without chromophore concentration changes). ϵ_n is the extinction coefficient of the chromophore n (in $L.mol^{-1}.cm^{-1}$). ΔC_n is the chromophore n concentration changes (in $mol.L^{-1}$). The spectral sensitivity of the detector i of the spectral acquisition device is represented by $D_i(\lambda)$ and $S(\lambda)$ is the normalized intensity spectrum of the light source. $L(\lambda)$ is the wavelength dependent mean optical path length of the travelled photons. Chromophores' concentration changes were obtained by matrix inversion once the matrix E has been calculated.

2.2 Liquid phantom

The liquid phantom (see Fig. 2 A) was composed of an homogeneous liquid mixture (1) that mimicked the grey matter and a transparent tube (2) that contained a absorbing liquid to mimic a periodic physiological event such as heartbeat. The tube made a vertical linear back and forth movement (from 2mm to $5mm$ from the surface of the liquid). This linear back and forth movement was performed in a period of 1s and was constantly repeated.

The liquid mixture was composed of a volume of $1.5L$ of phosphate buffered saline solution (PBS), 0.75g of Titanium dioxide powder (TiO_2) was used as the scattering material and 25g of a dilution of red liquid watercolour paint (Ecoline 350) and 25g of a dilution of blue liquid watercolour paint (Ecoline 551) were used as absorbing elements. The volumetric extinction spectra of the red and blue watercolour paint have been characterised by spectrophotometry. The red and blue dyes have an absorption pic at 562nm and at 628nm, respectively. The red and blue dilutions were prepared by mixing 1g of watercolour paint with 100g of scattering liquid (TiO_2 mixed with PBS). The transparent tube mentioned above was filled with the red dilution.

The liquid mixture was contained in a tank which was placed on a magnetic stirrer. This made it possible to homogeneously mix the liquid mixture when adding dilutions. Three 1-minute acquisitions were performed : a baseline, addition of 15g of the red dilution and addition of 15g of the blue dilution. Data were collected with a spectrometer *Ocean Optics USB 2000* (3) that collected light under the tube (2). Two optical fibres were placed under the tube and were used as a sourcedetector probe with a separation of 0.8cm. The massic concentration of TiO_2, red and blue dyes contained in the liquid phantom are indicated in table 1.

The optical path length of travelled photons at wavelength λ ($L(\lambda$ in $cm)$ between the source and the detector probes (see Fig. 2 A) can be obtained with the differential pathlength factor (DPF) and the source-detector separation d ($d = 0.8cm$):[9]

	C_{TiO_2} (%)	C_{Red} (%)	C_{Blue} (%)
Baseline	0.055	0.018	0.018
Red dilution addition	0.055	0.029	0.018
Blue dilution addition	0.055	0.029	0.029

Table 1. Massic concentration of TiO_2, red and blue dyes (expressed in %) contained in the liquid phantom.

$$L(\lambda) = DPF(\lambda).d = 0.5.\frac{3\mu_s^{'}(\lambda)}{\mu_a(\lambda}^{1/2}.d, \tag{4}$$

with $\mu_s^{'}$ (in cm^{-1}) the reduced scattering coefficient ($\mu_s^{'} = \mu_s.(1-g)$) and μ_a (in cm^{-1}) the absorption coefficient of the liquid mixture, see Fig. 1.

2.3 Numeric phantom

A homogeneous volume of grey matter has been modelled and we simulated the acquisition of the reflection spectra at its surface with a spectrometer (400nm to 1000nm) and a RGB camera (*BASLER acA2000-165uc*) using Monte Carlo simulations.[7] The mean path of travelled photons was also estimated. In this grey matter model, we incorporated the hemodynamic changes following a 10s neuronal stimulation and those due to heart-beat, see dotted lines in Fig. 2 C. Different shapes for ΔC_{HbO_2} and ΔC_{Hb} were voluntary chosen to evaluate crosstalks between the chromophores. A 30s acquisition (10Hz) was modelled under a homogeneous white light illumination, see Ref[8] for further details. Absorption and reduced scattering coefficients of the numeric phantom are represented in Fig. 1.

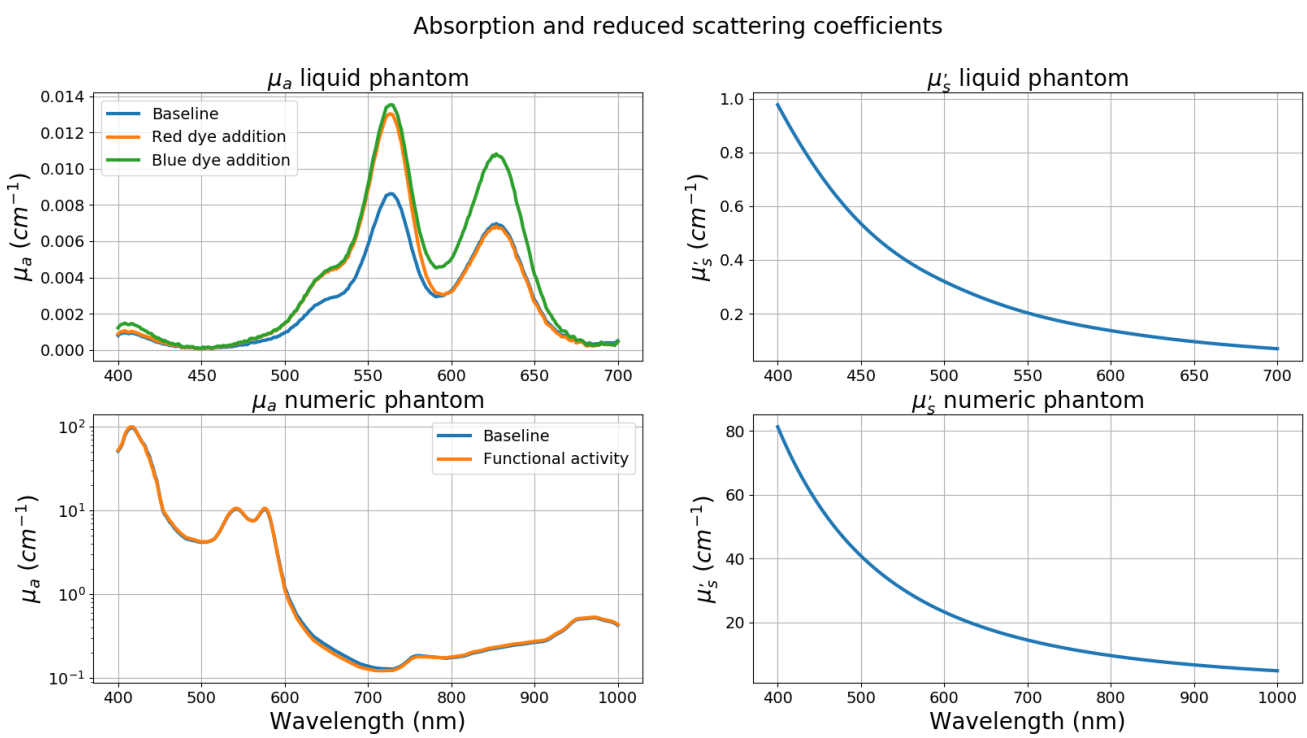

Figure 1. Absorption and reduced scattering coefficients of the liquid and numeric phantoms.

For both phantoms, concentration changes of Hb, HbO_2, red and blue dyes were calculated with the modified Beer-Lambert law.

3. RESULTS AND DISCUSSION

The concentration changes calculated with the modified Beer-Lambert law are represented in Fig. 2 B and C for both phantoms. With the liquid phantom, the USB-2000 can quantify ΔC_{Red} with low errors (9% average error) but cannot quantify ΔC_{Blue} precisely (59% average error). Errors were computed during the dyes addition periods (second and third acquisition periods). With the numeric phantom, the RGB camera can quantify ΔC_{Hb} with low errors (5.73% estimation error at $t = 10$s) but cannot quantify ΔC_{HbO_2} precisely (51% estimation error at $t = 10$s).

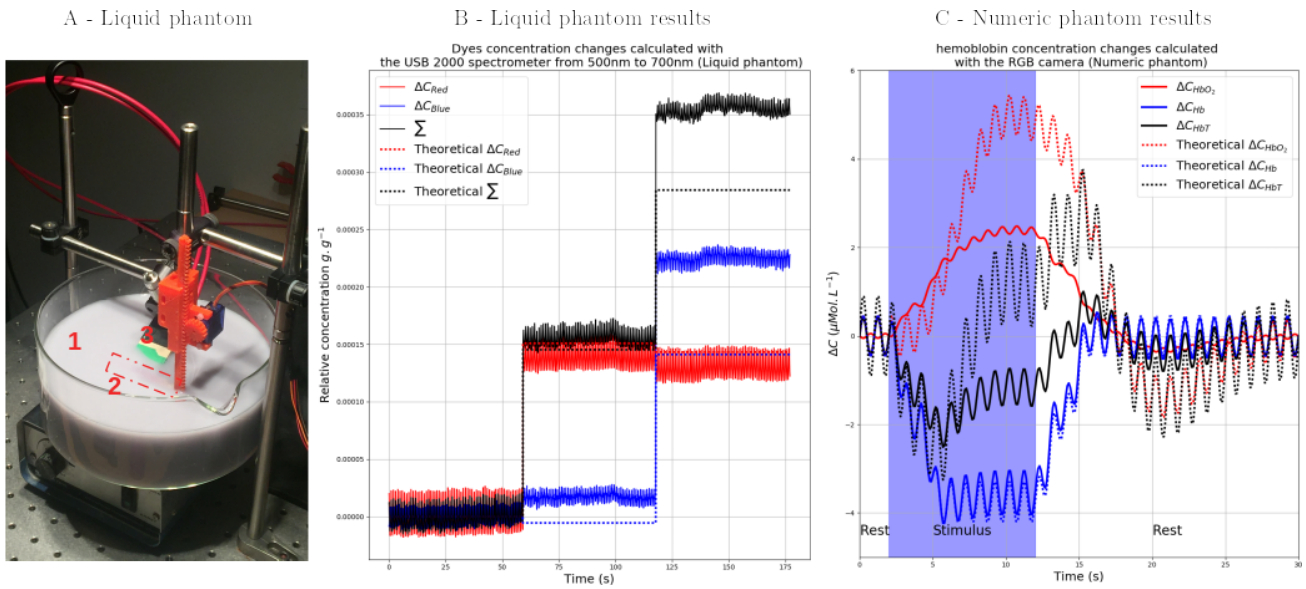

Figure 2. Numeric and liquid phantom used to model brain hemodynamics and concentration changes calculated with the modified Beer-Lambert law for both phantoms. (1) Liquid mixture. (2) Tube filled with red dilution. (3) Source-detector probe used to collect data with a spectrometer.

4. CONCLUSION

In this work, we presented the methodology to design a liquid and a numeric brain-simulating phantoms. These phantoms mimicked an exposed cerebral cortex as well as the slow concentration changes that occur after neuronal stimulation and the periodic changes due to heartbeat. These phantoms can be used to evaluate the robustness and the quantification performance of experimental setups. In future studies, the system to mimic the periodic heartbeat event can be improved. For example, a pump system could be used.

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